Lecture 17

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2026-11-04

Midterm 2

- Midterm 2 will be in-class next Wednesday, November 11.
- This week's lab will be a review for the code that you will need to know for the midterm.
- Next week's lab will be discussing an optional topic.
- Midterm 2 covers probability:
  - Chapters 13 - 17 in the text,
  - Lectures 9 - 15 and associated review exercises,
  - Homework Assignments 5-7, and
  - R material from lectures and labs related to probability and simulation.

R Code on Midterm 2

- Midterm 2 may ask for R code.
  - Calculate binomial or hypergeometric probabilities.
  - Simulate random variables.
  - Estimate probability, expected value, or standard error by simulation.
  - Use normal approximation to estimate probability.

Lecture 17

1. Reading Summary - Chapters 19 and 20
2. A Comparison of Survey Methods
3. Brexit
4. Five Thirty Eight and the 2022 Midterm Elections

Read Chapters 21 and 23 for next time.

Reading Summary - Chapter 19

Section 19.1: Introduction

- Definitions
  - **Population** - a group of individuals that are of interest
  - **Census** - a study of everyone in the population
  - **Sample** - a subset of the population that is more feasible to study
  - **Inference** - generalizing from the sample to the population
  - **Parameter** - a numerical fact about a population
  - **Statistic** - a number which can be computed from the sample

Section 19.1: Introduction Part 2

- Statistics are what we know.
  - We hope to use statistics to make inference about parameters.
  - Inference is accurate only if the sample is **representative** of the population.

Section 19.2: The Literary Digest Poll

- Historical example of a badly designed poll
- **Selection bias** - a systematic tendency to exclude a certain type of person from the sample
- If a selection procedure is biased, having a large sample doesn't help.
- **Non-response bias** - bias that is a result of people not responding to the survey
  - **Responder bias** - bias that is a result of people responding to the survey
- Satisfaction surveys are prone to responder and non-response bias.
- To determine if a sample is any good, examine the selection process

Section 19.3: The Year the Polls Elected Dewey

- Another historical example of a failed poll
- **Quota sampling** - hand-picking a sample to resemble the population with respect to important variables such as age, gender, etc. . .
  - NOTE: no randomization
- Quota sampling seems reasonable, but can do poorly.

Section 19.4: Using Chance in Survey Work

- **Probability methods** pick the sample using a chance mechanism.
- **Simple random sample**: a probability method in which every possible sample of a certain size has the same chance of being selected.
  - Think draws from a box without replacement.
- **(Multi-stage) cluster sampling**: First randomly select subgroups, then randomly select individuals from the subgroups.

Section 19.5: How Well do Probability Methods Work?

- Can work quite well.

Section 19.6: A Closer Look at the Gallup Poll

- Examines several details of polling
- Even with random sampling, difficulties remain.
  - Even after interviewing someone, we still don't know for sure what they will do on election day.
  - Non-voters
  - Undecided
  - Response or non-response bias

Section 19.7: Telephone Surveys

- How to randomly sample phone numbers? Two main options:
  - Choose randomly from phone book
  - Random digit dialing
- Each method has its own types of biases.

Section 19.8: Chance Error and Bias

- estimate = parameter + bias + chance error
- Usually bias is a larger concern than chance error.
- With a probability sample, we can quantify chance error mathematically.
- Bias is hard to detect and hard to quantify.

Reading Summary - Chapter 20

Section 20.1: Introduction

- Hypothetical conversation introducing chance variability
- Size of chance error mainly depends on size of sample, not size of population

Section 20.2: The Expected Value and the Standard Error of a Sample Percentage

- $\text{expected value} = \text{population percentage}$
- $\text{standard error} = \dfrac{\text{SD of box}}{\sqrt{n}} = \sqrt{\dfrac{p(1-p)}{n}}$

Note: These are formulas we covered in lecture 15.

Section 20.3: Using the Normal Curve

- Detailed example of using the normal curve
- Be careful to verify that the normal approximation is justified!

Section 20.4: The Correction Factor

- When drawing with replacement, the standard error depends only on the sample size, and not the population size.
- When the sample size is small compared to the population, there is essentially no difference between drawing with replacement and drawing without replacement.
- Conclusion: in most realistic situations, the size of the population doesn't matter.
- To be exact:

$$\text{SE without} = \text{SE with} \times \sqrt{\frac{N-n}{N-1}}.$$

Section 20.5: The Gallup Poll

- Reminder: only the size of the sample matters, not the size of the population.
  - That is why the Gallup poll can get good results with only a tiny fraction of the country.

A Comparison of Survey Methods

Convenience Sample

- A generic term we will use for any non-random sample
  - Polling people you encounter on the street would be a classic example.
  - Inviting everyone in class to fill out an online survey would be another example.
  - The big concern is that a convenience sample is not going to be representative.

Quota Sampling

- A specific type of non-random sample
- Requires that for certain variables (age, race, gender, etc) the sample is representative of the population.
- For other variables, the sample might not be representative at all.

Simple Random Sample

- Essentially drawing without replacement from entire population
- Requires everyone to be known in advance
- Rarely practical, or even possible

Cluster Sample

1) Divide the population into groups, often geographic.
2) Select some groups at random.
3) Sample the selected groups using:
   - Simple random sampling
   - Cluster sampling
     - Divide the selected groups into subgroups.
     - Select some subgroups at random.
   - Another sampling method

- A **multistage** sample has multiple rounds of cluster sampling.

Cluster Sample vs Simple Random Sample

- A cluster sample is much more feasible than a simple random sample.
- A cluster sample has a higher standard error than a simple random sample.

Stratified Sample

1) Divide (stratify) the population into groups, such as by age, race, gender, or other important variables.
2) Take a simple random sample of each group, proportional to the size of the group.

- The overall sample is therefore representative of the population in terms of the variables used in the stratification.
- Stratification requires that we know and have information on everyone in advance.
- Stratification lowers standard errors as compared to a simple random sample.

Quota Sampling vs. Stratified Sampling

- Both require that the sample is representative of the population for certain key variables such as age, race, gender, etc. . .
- However,
  - stratified sampling incorporates randomization by requiring a simple random sample within each group, while
  - quota sampling does not incorporate randomization.

Cluster Sampling vs. Stratified Sampling

- Both divide the population into subgroups.
- However,
  - cluster sampling only selects and samples from some subgroups, while
  - stratified sampling samples from all subgroups.

Brexit

Brexit Polls

- A June 23, 2016 poll of $4,772$ British citizens found that 52% supported remaining in the European Union, while 48% supported leaving the European Union.
  - See Poll Link
- The simple random sample formula for the standard error gives $\overline{0.52 * 0.48/4772} = 0.007$.
- Based on this standard error, it would seem highly unlikely that 50% or fewer British citizens support remaining.
  - $(0.50 - 0.52)/0.007 = -2.85$ standard errors

Brexit Polls Part 2

- The standard error in the previous slide is deceptive for two reasons:
  1) The sample was not a simple random sample.
     - The standard error will be higher than the standard error for a simple random sample.
  2) The standard error ignores bias.

Five Thirty Eight and the 2022 Midterm Elections

Five Thirty Eight

- ABC eliminated Five Thirty Eight in March 2025.
- Five Thirty Eight used data and statistical analysis to try to answer questions in politics, sports, and public health.
- It was known for the accuracy of its' elections forecasts.
- It forecasted elections by summarizing poll results.
  - If one poll provides information, then multiple polls provide more information.
- Five Thirty Eight considered uncertainty in a much more sophisticated way than most political pundits.

Five Thirty Eight and the 2022 Midterm Elections

- Five Thirty Eight created a model to forecast the results of the 2022 Midterm Elections.
- The model did a better job forecasting the election than most political pundits, who predicted a "red wave".
- The actual results were somewhat in the tail of what Five Thirty Eight predicted, however.
- The biggest difference between the forecast and what actually happened was in the Georgia Senate race.
- What happened?

Five Thirty Eight and Poll Aggregation - Poll Quality

- Some potential problems with poll aggregation
  - Aggregating garbage polls will yield garbage information, and polls are not always transparent about their methods.
  - Five Thirty Eight tried to account for this by giving letter grade ratings to pollsters and weighting the polls based on the ratings.
  - These ratings were based on the pollster performance in recent elections.
    - They did not consider pollster methodology.

Five Thirty Eight and Poll Aggregation - Poll Recency

- Five Thirty Eight weighted polls based on how close they were to the election.
- This was completely sensible.
- Partisan pollsters could use this to game the system by releasing polls right before the election.
  - Trafalgar Group is a Republican pollster who released lots of poll results right before the election.

Five Thirty Eight and Poll Aggregation - Herding

- You expect to see variation in polling results.
- Polling results that are overly consistent should raise a red flag.
  - Pollsters sometimes use the term **herding** to refer to this phenomenon.
- There was perhaps some evidence of herding in Nevada.
- There was more herding in the 2016 presidential election polls.

Review Questions

- See Review_questions17.pdf